Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Full-stack spin-qubit quantum computer development**			
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (✓),** Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), **Quantum (✓)**			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	*Other Semiconductor Devices*	*Software*	*System Integration*
Concept of Proposed Technology	Quantum computing holds the promise of major disruptions in high-performance computing. Based on quantum bits (qubits) that leverage the quantum mechanical properties of entanglement and superposition, quantum computing offers an exponential increase in computation power. If successful, this could provide breakthrough advancements in applications such as logistic optimization, the simulation of molecular systems, artificial intelligence, and encryption, with many new applications rapidly being discovered. Physical qubits come in different modalities that depend on the mode of exploitation of the underlying properties of quantum mechanical particles. In particular, spin qubits are based on electrons or holes confined in dots or defect centers in silicon or germanium semiconducting materials, respectively, either near the interface with an insulator or in Si/SiGe heterostructures. Spin qubits offer multiple advantages as quantum processing units (QPUs). These qubits have very long spin coherence times and have high potential for scalability, as spin qubits can leverage existing semiconductor manufacturing infrastructure for their production and may allow the use of existing VLSI-based circuit design software tools. In addition, spin qubits can operate at elevated temperatures above 1 K, which offers high compatibility with cryogenic electronics that operate at ~1.8 K, thereby reducing the scaling overhead. Furthermore, the relatively small physical size of spin qubits allows for high-density production and scalable chip-scale integration, reducing the overall footprint of a fully integrated quantum computer. This in turn offers the potential benefit of seamless integration with existing high-performance computing (HPC) clusters without the need for large supporting infrastructure.			

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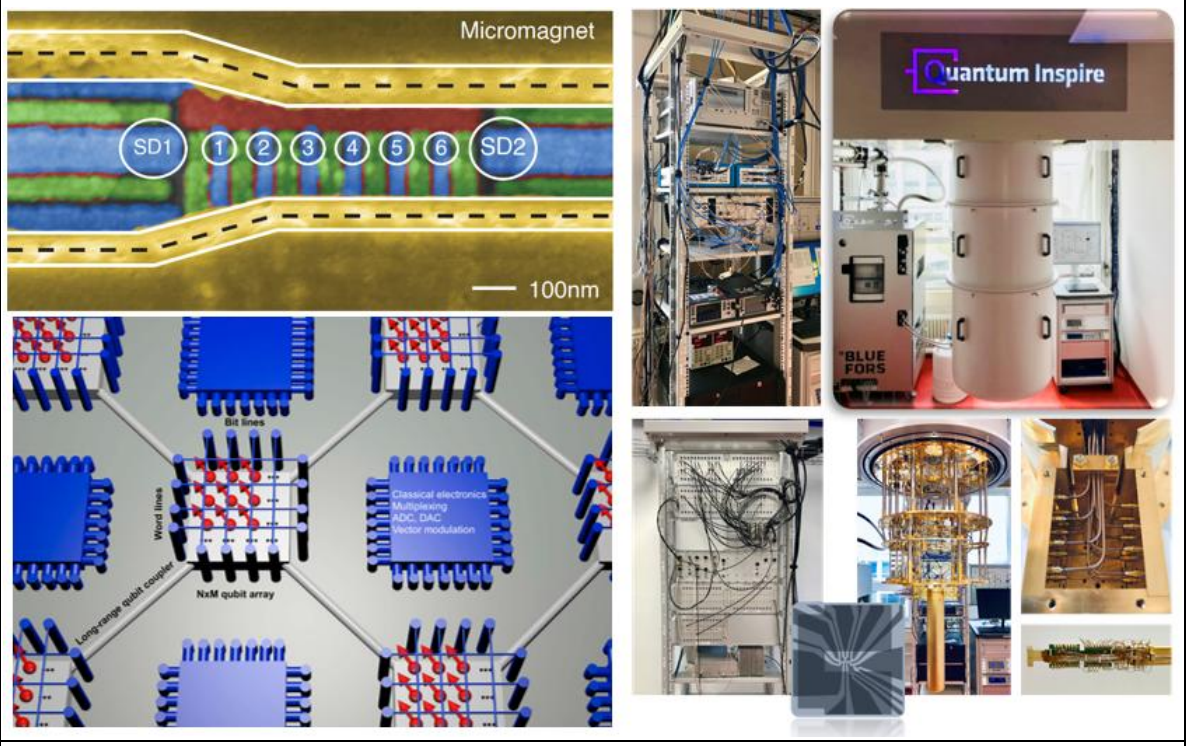
	The objective of this project is to develop a full-stack, spin-qubit–based quantum computer system and implement it in the form of a practically usable quantum cloud service. To achieve this goal, the project integrates end-to-end technologies ranging from hardware to software, as well as control and operational platforms for spin-qubit quantum computers. 
Rationale for Joint R&D Project	TNO Quantum is a leading European technology institute specializing in the research, development, and deployment of next-generation quantum hardware. Its expertise spans the lower level hardware stack, including the design and fabrication of advanced spin-qubit devices [1, 2], cryogenic measurement infrastructures, qubit control electronics, and calibrated low-level control software [3]. Through its state-of-the-art development facilities and deep scientific knowledge, TNO delivers quantum hardware that is engineered for stability, reproducibility, and scalable operation—key prerequisites for transitioning quantum processors from laboratory prototypes to cloud-accessible, industrial-grade systems. On the other hand, a highly innovative technology partner company focused on developing upper-stack software for quantum computing would be entailed. First, the partner company maintains a cloud-based platform service that provides user-friendly access to multiple instances of real quantum computers of different qubit implementation types (e.g. superconducting, trapped ions, etc.). Designed with their expertise on upper-to-lower stack circuit compilation, scheduling, and user end interfaces, the partner company's platform provides an easy-to-use workflow from software programming to execution on an appropriate modality of qubits. Second, the partner company also specializes in the research and development of software-level solutions with quantum computing. Their results range from interdisciplinary academic projects in topics such as drug discovery to applications in the financial sector. As such, the partner company should possess valuable technical expertise in key elements of the upper stack such as qubit-agnostic software abstractions and algorithm development. Through the modular, cloud-oriented software infrastructure and highly dedicated software researchers, the partner company accelerates the adoption of quantum computing by bridging the gap between complex device physics and user-end practical applications. This proposal strategically brings together TNO's hardware excellence and the partner company's software innovation to create a unified end-to-end quantum computing service. TNO provides the high-fidelity qubit devices, calibrated control stack, and operational know-how required to maintain robust quantum processors. The partner company complements this with a powerful, user-friendly software ecosystem that exposes the computational capabilities of those processors to developers, researchers, and industrial customers. By tightly coupling hardware-level performance with

	sophisticated software orchestration, the collaboration offers a uniquely integrated solution: a quantum computing service that is technically advanced, commercially scalable, and immediately accessible to end users. From a market perspective, the combination positions TNO and the partner company's as a strong, differentiated partnership—one that not only pushes the boundaries of quantum technology but also lowers the barrier for real-world adoption. Customers benefit from cutting-edge European hardware innovation paired with a modern software environment that supports rapid experimentation, benchmarking, and application development. The result is a compelling value proposition: a dependable, high-performance quantum computing platform delivered through a seamless and professional service model.
Key Words	Provide key terms that describe the proposed technology and its main features (min 3, max 5) Semiconductor, spin qubit, quantum algorithm performance, quantum benchmarking, quantum cloud platform

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	A summary of the objectives is as follows: 1. Developing a semiconductor-process-based spin-qubit quantum processor 2. Establishing a cryogenic system capable of stable operation in ultra-low-temperature environments. 3. Developing quantum circuit optimization software 4. Developing task-based benchmarking and performance evaluation tools The project focuses on developing and improving the semiconductor-based spin-qubit quantum processor (QPU) and a cryogenic system for stable ultra-low-temperature operation. It also aims to establish core spin-qubit technologies such as tuning, calibration, noise characterization, and stabilization for a reliable quantum computing environment. On the software side, the project develops quantum circuit optimization tailored to spin-qubit constraints – coupling topology, gate sets, and error characteristics – enabling efficient compilation, mapping, and scheduling. Additionally, it creates benchmarking tools for algorithm-level performance evaluation, providing validation of practical spin-qubit quantum computing. Ultimately, the project integrates these hardware and software technologies into a quantum cloud platform, enabling users to access spin-qubit quantum computers and perform quantum circuit design, execution, and analysis. Through this platform, the project aims to demonstrate the practical applicability of spin-qubit-based quantum computing to researchers and industrial users, while establishing a solid technological foundation for commercializable next-generation quantum computing services.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* ***Hardware (TNO)*** • Training and education programs to cultivate specialized personnel with enhanced capabilities in control of silicon-based spin-qubit devices • Development of design technologies for spin-qubit systems that ensure precise control and stable operation such as design of micromagnet magnetic field gradients for implementing electric-dipole spin resonance (EDSR) • Development of qubit fabrication process technologies that satisfy ultra-high performance and long coherence times and improvements of device design (device supplied by TNO)			

and/or validating materials and devices supplied by sub-contractors/suppliers.

- Design and fabrication of cryogenic packaging for qubit devices
- Development of high-speed, high-reliability quantum gate calibration technologies
  - Optimization of gate parameters for qubit operation, initialization, and readout
  - Optimization of crosstalk issues in multi-qubit gates
- Performance evaluation of fabricated QPUs through execution of benchmark programs on the following parameters:
  - Physical qubits 2-8 qubits
  - Coherence time (T2)
  - Coherence time (T2 Hahn)
  - 1Q gate fidelity
  - 2Q gate fidelity
  - Qubit initialization success rate

Software and Integration with a Service Platform (partner company) [WIP]

Upper-Stack Software (Quantum AI Team)

- Spin-qubit-based quantum circuit optimization toolkit
  - In-house theoretical and software-level tools for optimizing input quantum circuits and algorithms at the software stage (e.g., algebraic-property-based optimizations such as commutativity and ZX-calculus-based simplification [4], geometry-aware qubit routing, and noise-adaptive cost modeling)
  - Customized algorithms for each optimization objective (e.g. reducing gate counts, reduction of noise sensitivity) compatible to the hardware
  - Software tools that convert user-specified quantum circuits/programs into efficient and stable forms on the hardware

- Spin-qubit-based algorithm performance benchmarks
  - Key small-scale algorithms (e.g., variational updates, SWAP operations, preparation of two-qubit entangled states) to benchmark software performance beyond hardware specifications.
  - Representative algorithms for various complexity levels, optimized and simplified with the previously developed optimization toolkit.
  - Development of benchmark metrics and metric evaluation software to validate the performance of user-requested quantum circuits/programs during execution

Cloud Platform Integration (Quantum Platform Team)

- Infrastructure development for spin-qubit quantum computer-cloud platform integration
  - A system infrastructure to connect the spin-qubit quantum computer with a service-oriented quantum cloud platform after hardware deployment
  - A system architecture and hardware/network specifications to ensure stable connectivity between the quantum computer and the cloud
  - A Quantum job management system and interfaces to manage the end-to-end workflow from development and execution to result delivery
  - Resource management, queuing, and access control architectures that support multi-user and multi-job environments

- System Software Development for Spin-Qubit-Based Systems
  - A real-time monitoring system for key parameters reflecting spin-qubit characteristics (e.g., coherence times, frequencies, gate fidelities, readout errors)
  - Performance dashboards and visualization interfaces that allow intuitive understanding of quantum computer status and performance
  - A workflow framework that integrates circuit generation, compilation, execution, and result analysis

- Integration with a Web-Based Quantum Computing Service Platform
  - A web-based user interface (UI) that enables remote access to spin-qubit quantum

	computers, even for users without specialized expertise ○ An integrated user access system that supports quantum job submission, execution status monitoring, and result analysis ○ A comprehensive development environment supporting the full lifecycle—quantum algorithm design, quantum circuit programming, compilation, execution, testing, and debugging—tailored to spin-qubit quantum computers ○ Hardware-aware compilation, mapping, and optimization functionalities ● Development of Tutorials and Hands-On Training Content ○ Service-oriented training content covering spin-qubit quantum computer usage, circuit execution procedures, and result interpretation ○ Hands-on examples and guides based on well-known quantum algorithms and notable application cases
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* ***Hardware (TNO)*** **Year 1:** ● Renting out a device and making quantum hardware facilities available, including a dilution refrigerator, control hardware, and low-level control software. ● Providing high level training in measurement, tuning, and calibration of spin-qubit quantum processors. **Year 2:** ● Development of spin qubit devices that are either: ○ manufactured by TNO ○ or by an external sub-contractor/suppliers ● Improvement of device design (if manufactured by TNO) or development of material benchmarking tools (if materials and devices are used that are provided by external suppliers) ● Improvement of spin-based quantum computers to include up to 8 qubits. ● Optimization of micromagnetic fields for improved qubit control and decreased crosstalk. ***Software and Integration with a Service Platform (partner company) [WIP]*** **Year 1:** <u>Upper-Stack Software (partner company)</u> ● Training of partner company's personnel in spin qubit mechanics and architecture. ● Familiarization with TNO's existing hardware, synchronized with TNO's updates. ● Identification of appropriate benchmarks and internal optimization targets in the context of TNO's spin qubit hardware, such as: ○ Benchmarks: Error propagation rate, execution time, effective probability of successful computation, etc. ○ Internal optimization targets: Gate count and circuit depth, noise sensitivity of states and gates, etc. ● Research of compatible existing theoretical/software methods in: ○ Circuit optimization; ○ Performance benchmarking. ● Invention of new necessary methods. ● Compilation of software development kit (SDK) implementing identified methods. ● Design of an integration architecture between TNO's spin-qubit quantum computer and a cloud service with hardware/network specifications **Year 2:** ● Continued development of SDK. ● Realization of SDK functions on TNO's spin qubit hardware. ● Production of numerical results validating performance using developed methods. ● Further refinements & de-bugging – setting up an environment (shared between TNO and partner company) for continued long-term development. ● Development of a cloud-based quantum development platform integrated with a spin-qubit

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	quantum computer ● Development of a cloud platform for development with a spin-qubit quantum computer ○ Construction of monitoring systems and service dashboards ○ Provision of an end-to-end quantum development environment ○ Deployment of cloud infrastructure and system-level validation of spin-qubit hardware integration ● Cloud service testing and utilization ○ Testing and validation of representative quantum algorithms on the spin-qubit quantum computer ○ Development of software-level tutorials and hands-on training content

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* TNO will be responsible for hardware and low-level control software, including the complete measurement infrastructure as elucidated in-detail above.
Expected R&R of Korean SME Partner	Partner company will be responsible for the development of the upper-level software stack for TNO's spin-qubit quantum computer. The software will cover aspects including, but not limited to, circuit optimization and performance benchmarking. Partner company will also integrate the quantum system with a cloud-based service platform, providing platform management and web-based development environments to enable efficient and accessible quantum computing services.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* **1. Technological Achievements** ***Hardware Innovations:*** ● Development of a full-stack, spin-qubit–based quantum computer system, integrating hardware, software, and operational platforms. ● Creation of semiconductor-process-based spin-qubit quantum processors (QPUs) with advanced cryogenic systems for stable ultra-low-temperature operation. ● Implementation of core technologies for spin qubits, including qubit tuning, calibration, noise characterization, and stabilization, ensuring repeatable and reliable quantum computing. ***Software and Optimization:*** ● Development of quantum circuit optimization tools tailored for spin-qubit architectures, addressing constraints like coupling topology, gate sets, and error characteristics. ● Advanced compilation, mapping, and scheduling techniques to maximize execution efficiency and minimize errors. ● Creation of benchmarking and performance evaluation tools for quantitative analysis from an algorithmic perspective. ● Integration of hardware and software into a quantum cloud platform, enabling remote access and practical use for researchers and industry. **2. Commercialization Potential** ***Phased Roadmap:*** ● Phase 1: Secure core spin-qubit QPU technologies and demonstrate service integration with quantum cloud platforms. ● Phase 2: Develop commercial prototypes and advance cloud services for high-potential domains (finance, drug discovery, new materials, optimization). ● Phase 3: Mass production of commercial-grade quantum computers and global expansion through B2B services and partnerships. ***Business Model:*** ● Hardware supply of spin-qubit QPU–based quantum computing systems. ● Quantum Computing as a Service (QCaaS) via cloud platforms. ● Provision of development tools for algorithm optimization and validation. ● Client-specific consulting services. ***Service Deployment:*** ● Operation of a spin-qubit–based full-stack quantum computing system as a cloud service. ● Domestic deployment to foster a quantum computing ecosystem supporting diverse architectures. ● Establishment of platforms for quantum algorithm-based services and commercialization. ● Creation of a technology–demand feedback loop through user-centric service operation. **3. Broader Industrial and Economic Impact** ***Industry Transformation:*** ● Acceleration of quantum computing adoption in key sectors such as finance, pharmaceuticals, materials science, and logistics. ● Enablement of new computational capabilities for complex problem-solving, optimization, and simulation tasks previously infeasible with classical computers. ***Economic Growth:*** ● Creation of new business opportunities in hardware, software, and cloud-based quantum

	services. ● Fostering a skilled workforce through training and education programs in quantum technologies. ● Establishment of a robust quantum technology ecosystem, driving innovation and competitiveness on a global scale. ***Societal Benefits:*** ● Democratization of quantum computing access via cloud platforms, supporting both expert and non-expert users. ● Advancement of scientific research and industrial applications, contributing to technological progress and societal well-being. **In summary:** This joint R&D project is expected to deliver significant technological advancements in spin-qubit quantum computing, establish a scalable commercialization pathway, and generate broad industrial and economic benefits by enabling new applications, fostering innovation, and supporting the growth of a quantum technology ecosystem

6. References

[1] Philips et al.,Universal control of a six-qubit quantum processor in silicon, Nature 609, 919 (2022)

[2] Vandersypen et al., Interfacing spin qubits in quantum dots and donors—hot, dense, and coherent, npj Quantum Information 3, 34 (2017)

[3] Last et al.,Quantum Inspire: QuTech's platform for co-development and collaboration in quantum computing, Proceedings 11324 (2020)

[4] J. van de Wetering. "ZX-calculus for the working quantum computer scientist." arXiv preprint arXiv:2012.13966 (2020).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization